

Portfolio optimization problems in different risk measures using genetic algorithm

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Abstract(1/2)

This paper introduces a heuristic approach to portfolio optimization problems in different risk measures by employing genetic algorithm(GA) and compares its performance to mean-variance model in **cardinality constrained efficient frontier**. To achieve this objective, we collected three different risk measures base upon **mean-variance** by Markowitz: **semi-variance**, **mean absolute deviation** and **variance with skewness**.

Abstract(2/2)

We show that these portfolio optimization problems can now be solved by genetic algorithm if mean-variance, semi-variance, mean absolute deviation and variance with skewness are used as the measures of risk. The robustness of our heuristic method is verified by three data sets collected from main financial markets. The empirical results also show that the **investors should include only one-third of total assets into the portfolio** which outperforms than those contained more assets.

Harry Markowitz(1927~Now)



- Nobel Memorial Prize in Economic Sciences(1990)
- Known for
Modern Portfolio Theory
Efficient/ Markowitz Frontier
Sparse Matrix Methods
Simsript

Markowitz 建立數學模型以說明在固定的期望報酬之下，如何去選取風險最小的投資組合 (Mean Variance Model) 。

The expect returns of portfolio

$$E(R_p) = \sum_{j=1}^N w_j \cdot E(R_j) ,$$

- $E(.)$ is the expect value
- w_j is the proportion of the portfolio held in asset j

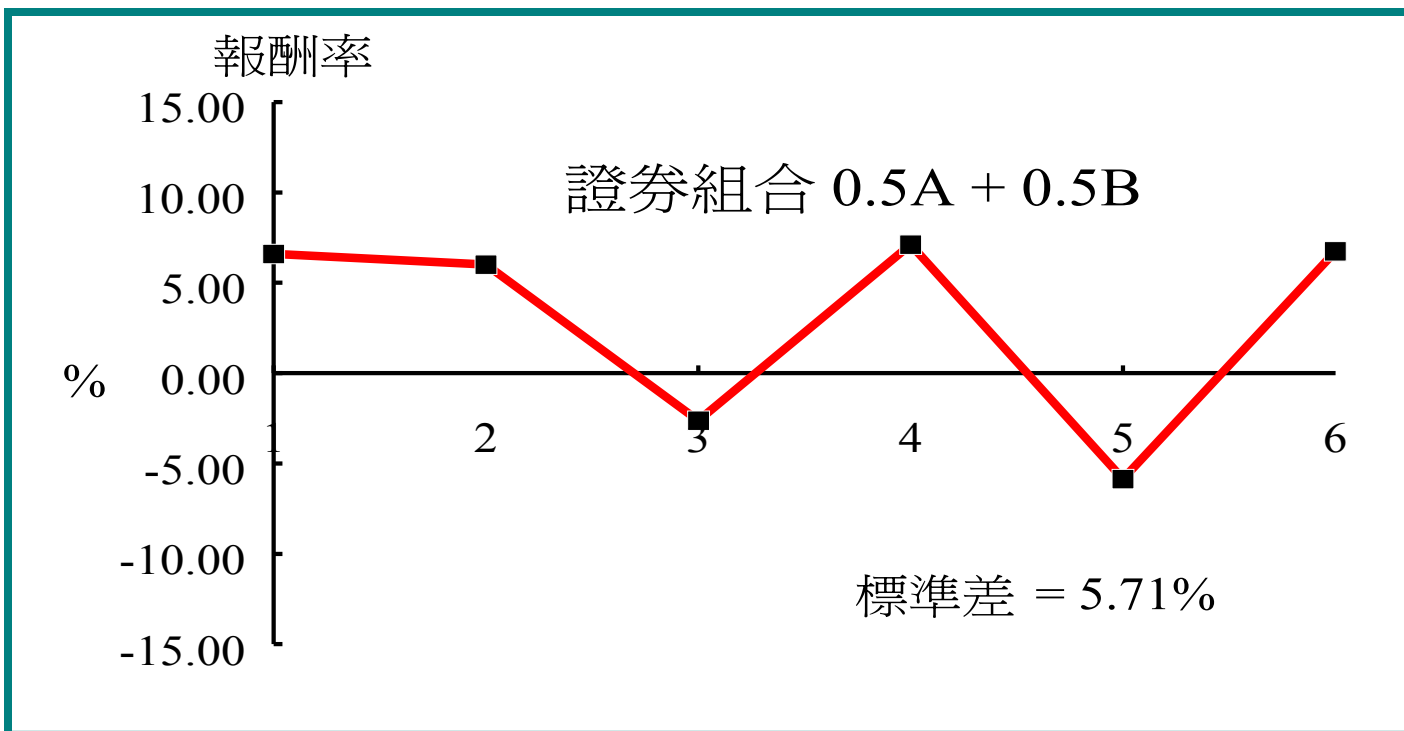
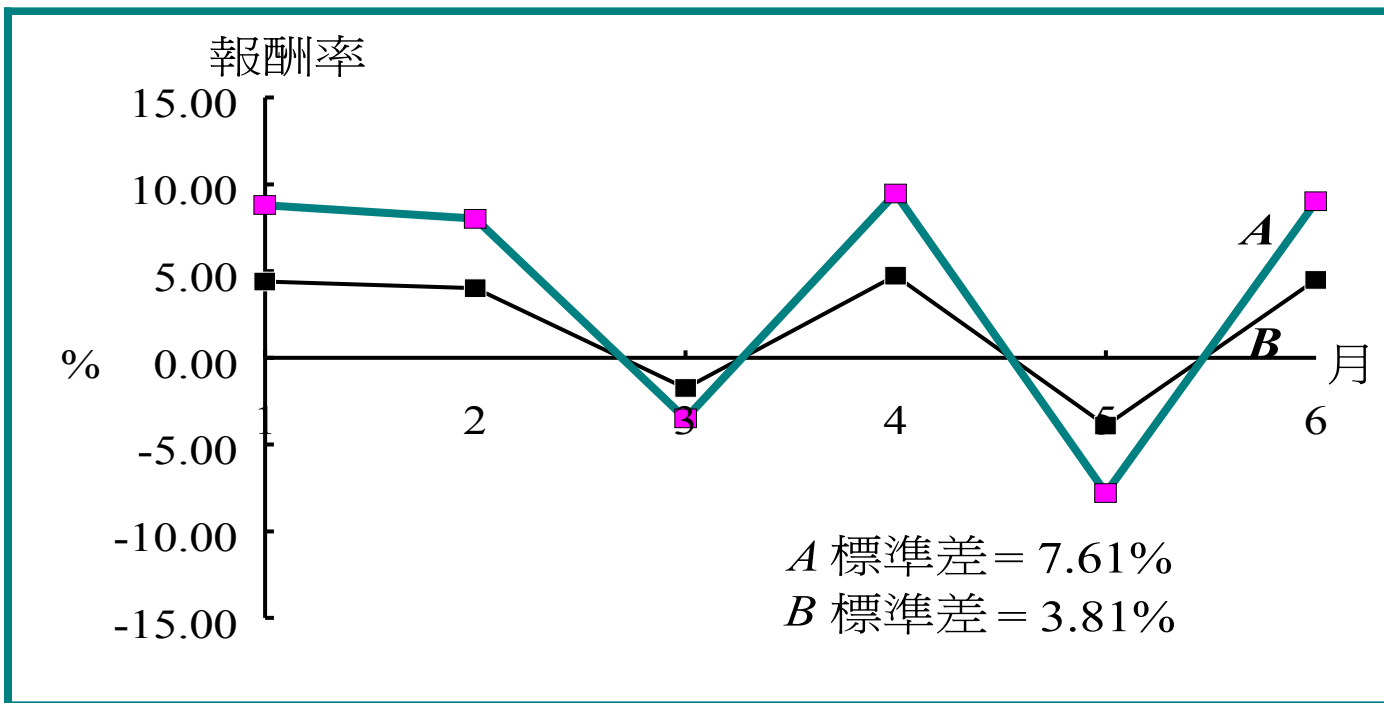
Example :

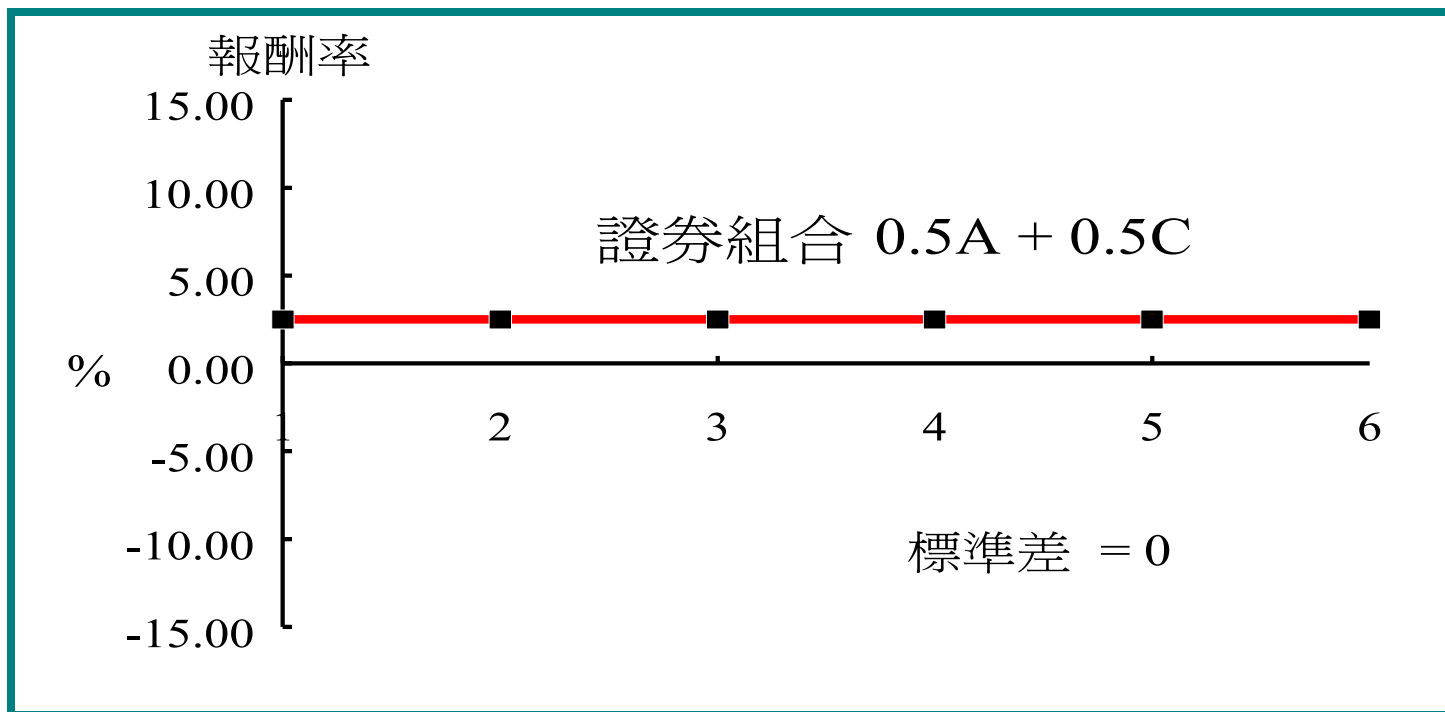
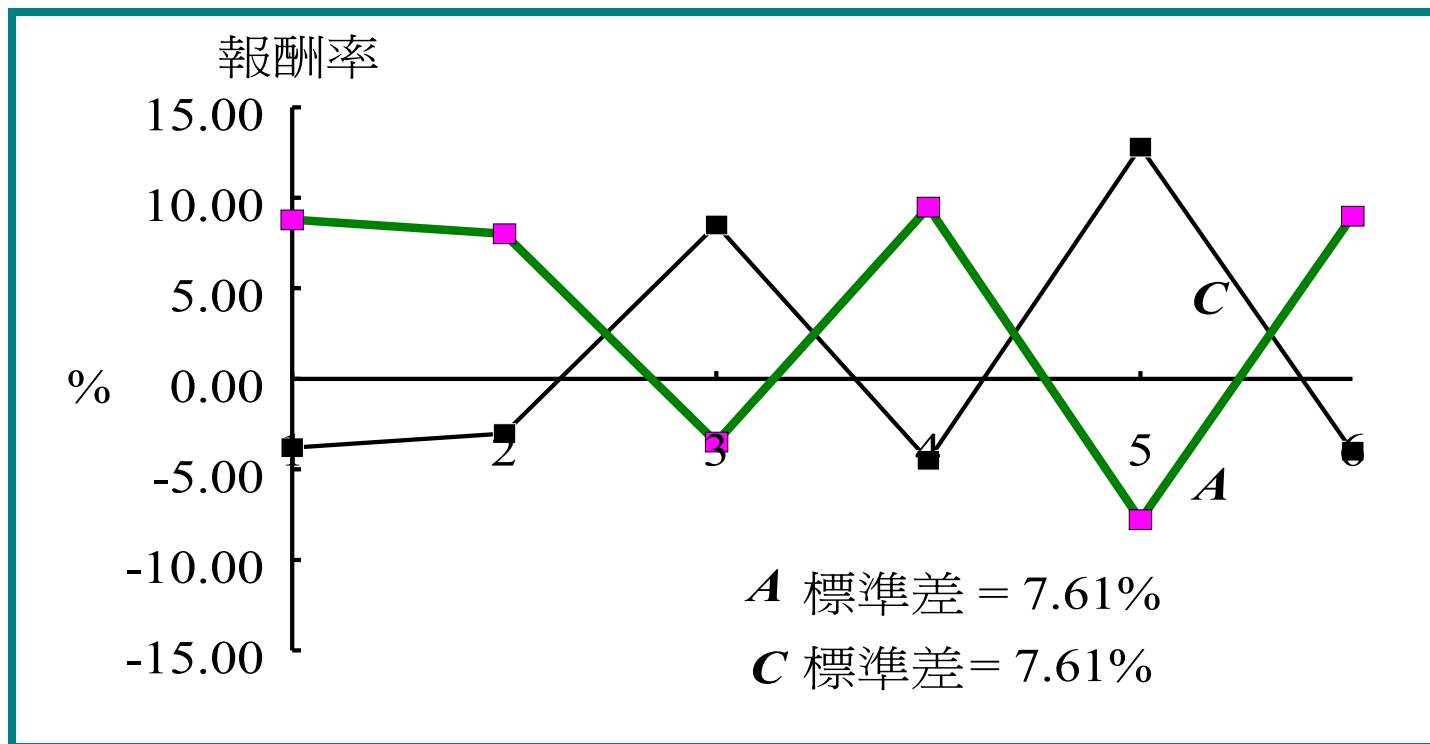
年初購買同樣金額的 X 、 Y 、 Z 三種股票，年底賣出，利率各為 6% ， 6.2% 、 6.4% ，則期望報酬率為何？

$$\begin{aligned} \text{答： } E(R_p) &= \frac{1}{N} \sum_{j=1}^N E(R_j) = \frac{1}{N} \times (6\% + 6.2\% + 6.4\%) \\ &= \underline{6.2\%} . \end{aligned}$$

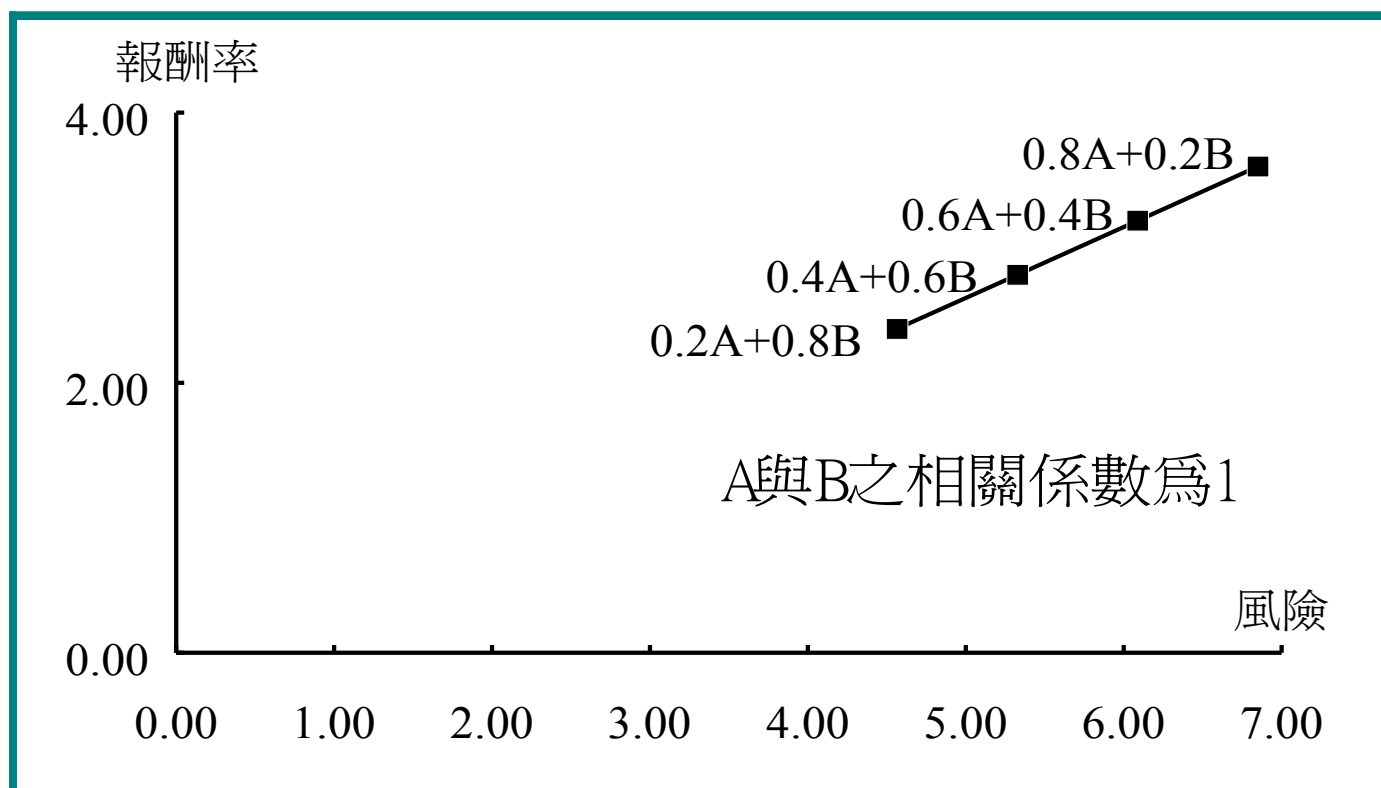
The risk of portfolio

個 股			%	證 券 組 合	
月	A	B	C	AB	AC
1	8.80	4.40	-3.80	6.60	2.50
2	8.02	4.01	-3.02	6.02	2.50
3	-3.50	-1.75	8.50	-2.63	2.50
4	9.48	4.74	-4.48	7.11	2.50
5	-7.80	-3.90	12.80	-5.85	2.50
6	9.00	4.50	-4.00	6.75	2.50
均數	4.00	2.00	1.00	3.00	2.50
標準差	7.61	3.81	7.61	5.71	0.00

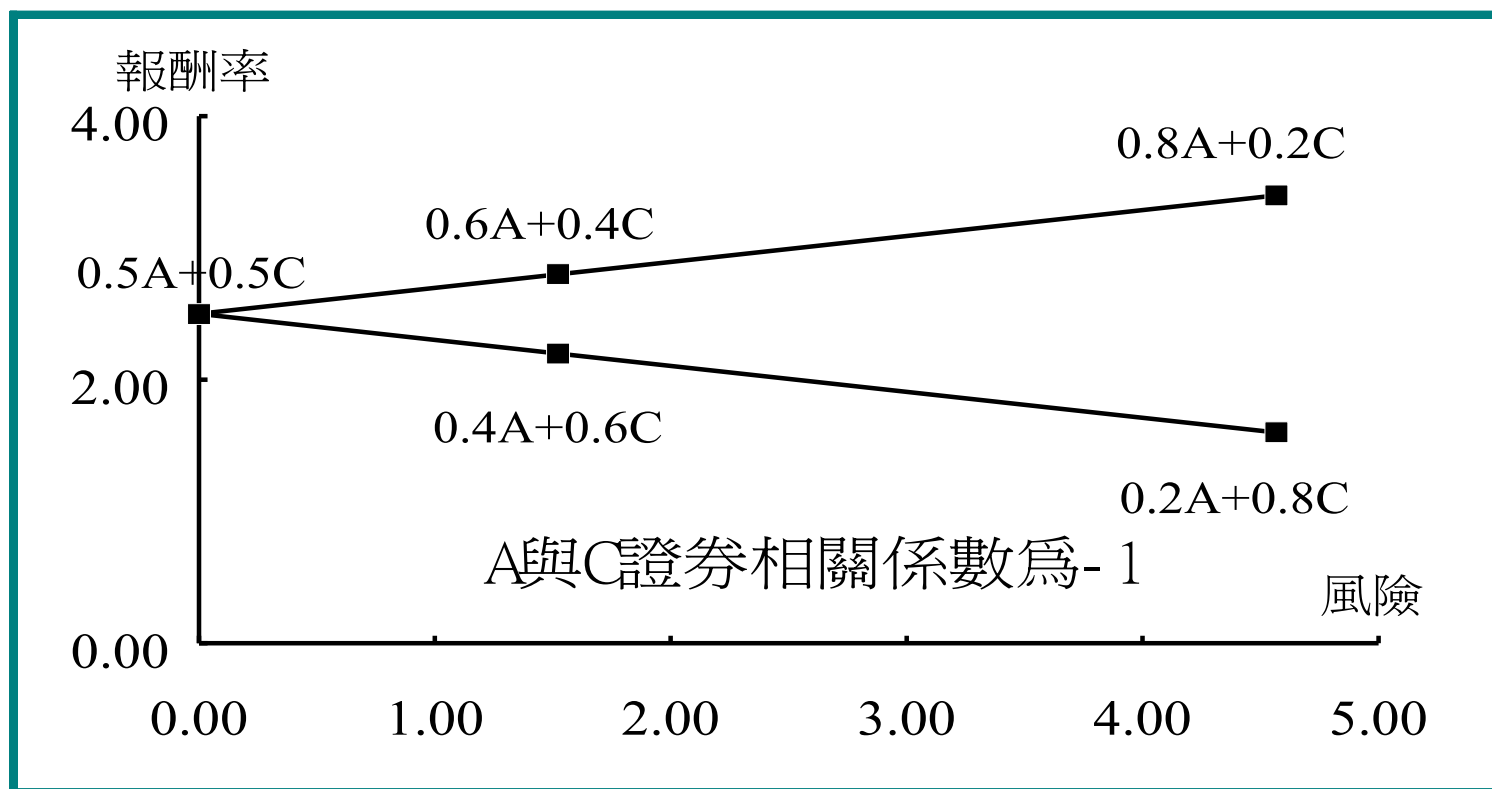




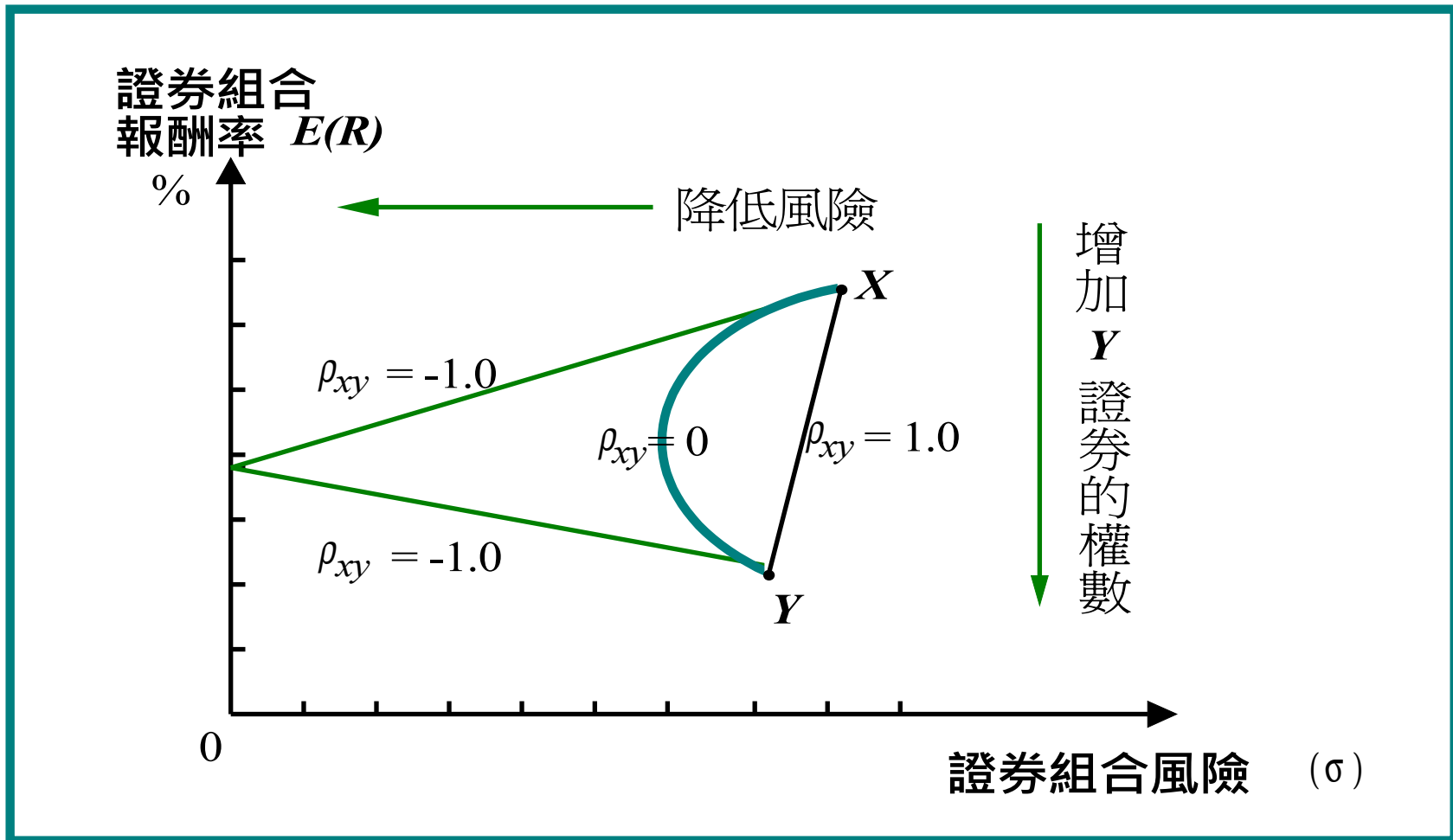
個股			證券組合				%
月	<i>A</i>	<i>B</i>	$0.8A+0.2B$	$0.6A+0.4B$	$0.4A+0.6B$	$0.2A+0.8B$	
1	8.80	4.40	7.92	7.04	6.16	5.28	
2	8.02	4.01	7.22	6.42	5.61	4.81	
3	-3.50	-1.75	-3.15	-2.80	-2.45	-2.10	
4	9.48	4.74	8.53	7.58	6.64	5.69	
5	-7.80	-3.90	-7.02	-6.24	-5.46	-4.68	
6	9.00	4.50	8.10	7.20	6.30	5.40	
均數	4.00	2.00	3.60	3.20	2.80	2.40	
標準差	7.61	3.81	6.85	6.09	5.33	4.57	



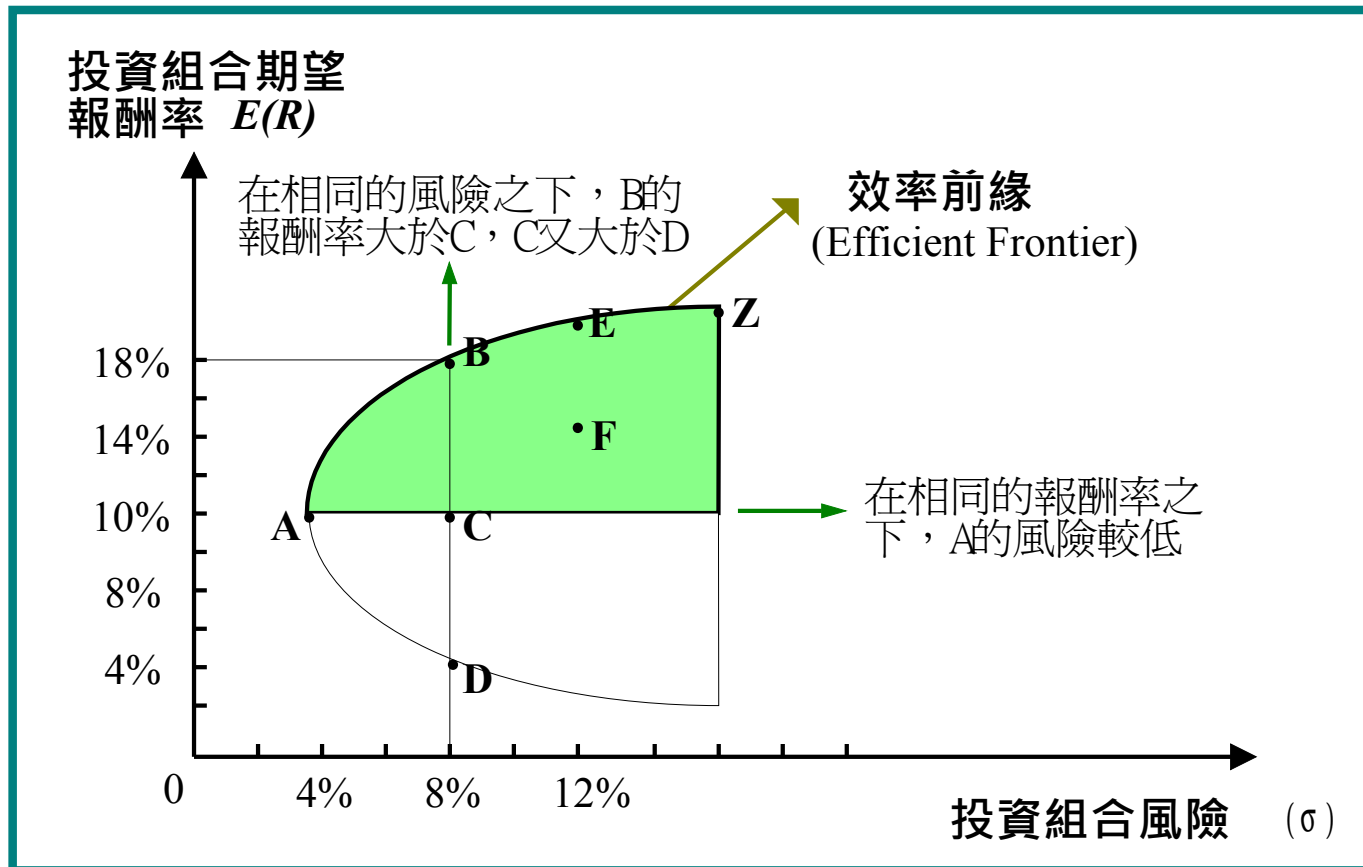
個股			證券組合				%
月	A	C	$0.8A+0.2C$	$0.6A+0.4C$	$0.4A+0.6C$	$0.2A+0.8C$	
1	8.80	-3.80	6.28	3.76	1.24	-1.28	
2	8.02	-3.02	5.81	3.60	1.40	-0.81	
3	-3.50	8.50	-1.10	1.30	3.70	6.10	
4	9.48	-4.48	6.69	3.90	1.10	-1.69	
5	-7.80	12.80	-3.68	0.44	4.56	8.68	
6	9.00	-4.00	6.40	3.80	1.20	-1.40	
均數	4.00	1.00	3.40	2.80	2.20	1.60	
標準差	7.61	7.61	4.57	1.52	1.52	4.57	



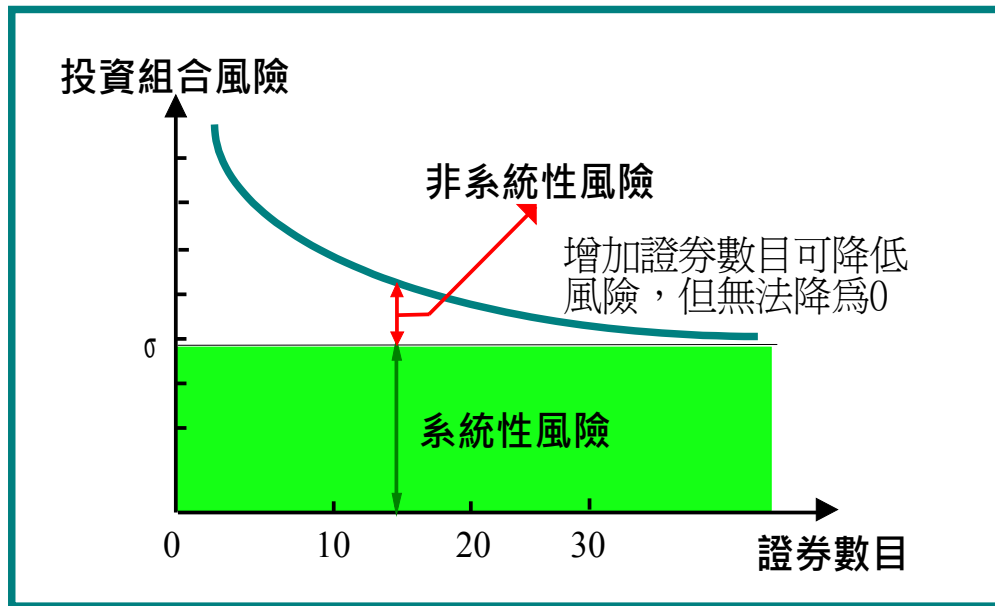
當證券的相關係數越低時，可分散風險的能力越高



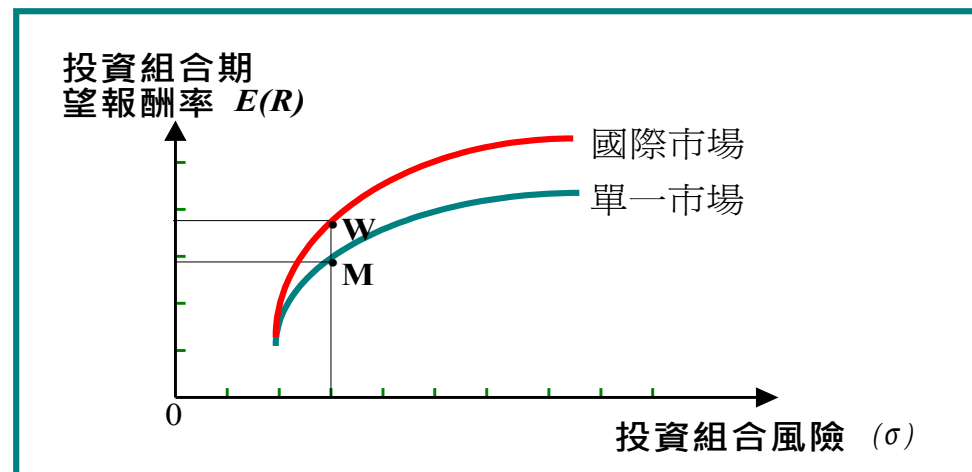
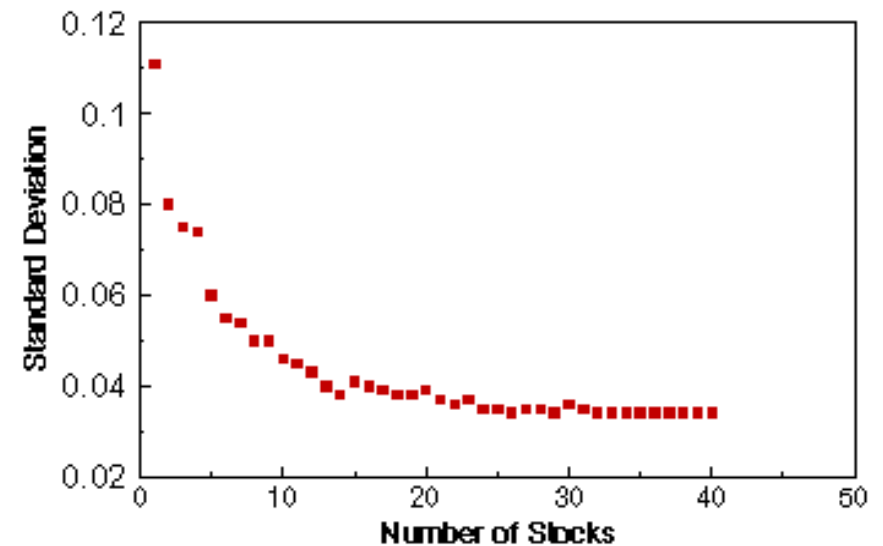
Efficient frontier



- A 與 C 報酬率相等，但 A 的標準差較小，所以 A 較有效率
- B 與 C 標準差相等，但 B 的報酬率較高，所以 B 較有效率
- C 與 D 標準差相等，但 C 的報酬率較高，所以 C 較有效率



Standard Deviation of Portfolio Return as a Function of Number of Stocks in Portfolio
From Fama (1976)



- The portfolio selection problem is to find all the efficient portfolios along this frontier.

Mean-variance model

$$\text{Minimize} \quad \lambda \left[\sum_{i=1}^N \sum_{j=1}^N w_i w_j \sigma_{ij} \right] - (1 - \lambda) \left[\sum_{i=1}^N w_i \mu_i \right] \quad (5)$$

$$\text{Subject to} \quad \sum_{i=1}^N w_i = 1 \quad (6)$$

$$\sum_{i=1}^N z_i = K \quad (7)$$

$$\varepsilon_i z_i \leq w_i \leq \delta_i z_i, \quad i = 1, \dots, N \quad (8)$$

$$z_i \in \{0, 1\}, \quad i = 1, \dots, N \quad (9)$$

N is the number of assets available;

w_i is the proportion ($0 \leq w_i \leq 1$) of the portfolio held in assets i ($i = 1, \dots, N$);

μ_i is the expected return of asset i ($i = 1, \dots, N$);

σ_{ij} is the covariance between assets i and j ($i = 1, \dots, N$; $j = 1, \dots, N$).

K is the desired number of assets in the portfolio;

ε_i is the minimum proportion that must be held of asset i ($i = 1, \dots, N$) if any of assets i is held;

δ_i is the maximum proportion that can be held of asset i ($i = 1, \dots, N$) if any of assets i is held;

$z_i = 1$ if any of asset i ($i = 1, \dots, N$) is held
 $= 0$ otherwise.

$\lambda = 0$: maximum expected return

$\lambda = 1$: minimum risk

$0 < \lambda < 1$: trade-off between risk and return

- It is a quadratic programming problem

It based upon the assumptions:

- Investor is risk averse
- The rate of return is normal distribution

Semi-variance model

$$w_i = v_{iT} x_i / C_{cash}, \quad i = 1, \dots, N \quad (10)$$

$$\sum_{i=1}^N z_i = K \quad (17)$$

$$\varepsilon_i z_i \leq w_i \leq \delta_i z_i, \quad i = 1, \dots, N \quad (18)$$

$$\sum_{i=1}^N w_i = 1 \quad (19)$$

$$r_t = \log_e \left[\left(\sum_{i=1}^N w_i v_{it} / v_{iT} \right) / \left(\sum_{i=1}^N w_i v_{i,t-1} / v_{iT} \right) \right], \quad t = 1, \dots, T \quad (20)$$

$$w_i \geq 0, \quad i = 1, \dots, N \quad (21)$$

$$z_i \in \{0, 1\}, \quad i = 1, \dots, N \quad (22)$$

$$\bar{r} = \sum_{t=1}^T r_t / T \quad (23)$$

T be such that we have observed historical values for stocks over the time period $0, 1, 2, \dots, T$;

v_{it} be the value of one unit of stock i ($i = 1, \dots, N$) at time t ($t = 0, \dots, T$);

C_{cash} be the cash available to invest in the portfolio;

x_i be the number of units of stock i ($i = 1, \dots, N$) that we choose to hold in the portfolio;

$z_i = 1$ if any of stock i ($i = 1, \dots, N$) is held in the portfolio
 $= 0$ otherwise.

w_i is the proportion of C_{cash} that is invested at time T in stock i ($i = 1, \dots, N$);

r_t is the single period continuous time return given by the portfolio at time t ($t = 1, \dots, T$).

$$\text{Minimize } \lambda \left[\sum_{t=1; r_t < \bar{r}}^T (r_t - \bar{r})^2 / T \right] - (1 - \lambda) \bar{r} \quad (25)$$

Subject to Eqs. (17)–(23)

$\lambda = 0$: maximum expected return

$\lambda = 1$: minimum risk

$0 < \lambda < 1$: trade-off between risk and return

股價下跌時視為風險，股價上漲時則不計算為風險

- still a quadratic programming problem

Mean absolute deviation model

$$\begin{aligned} \text{Minimize} \quad & \lambda \left[\sum_{t=1}^T |r_t - \bar{r}|/T \right] - (1 - \lambda)\bar{r} \\ \text{Subject to Eqs.} \quad & (17) - (23) \end{aligned} \tag{26}$$

- It can be solved via **linear programming**.
- The model generates similar results to the Markowitz mean-variance model.

Variance with skewness

The distribution of individual asset returns tend to exhibit a higher probability of extreme values than is consistent with normality.

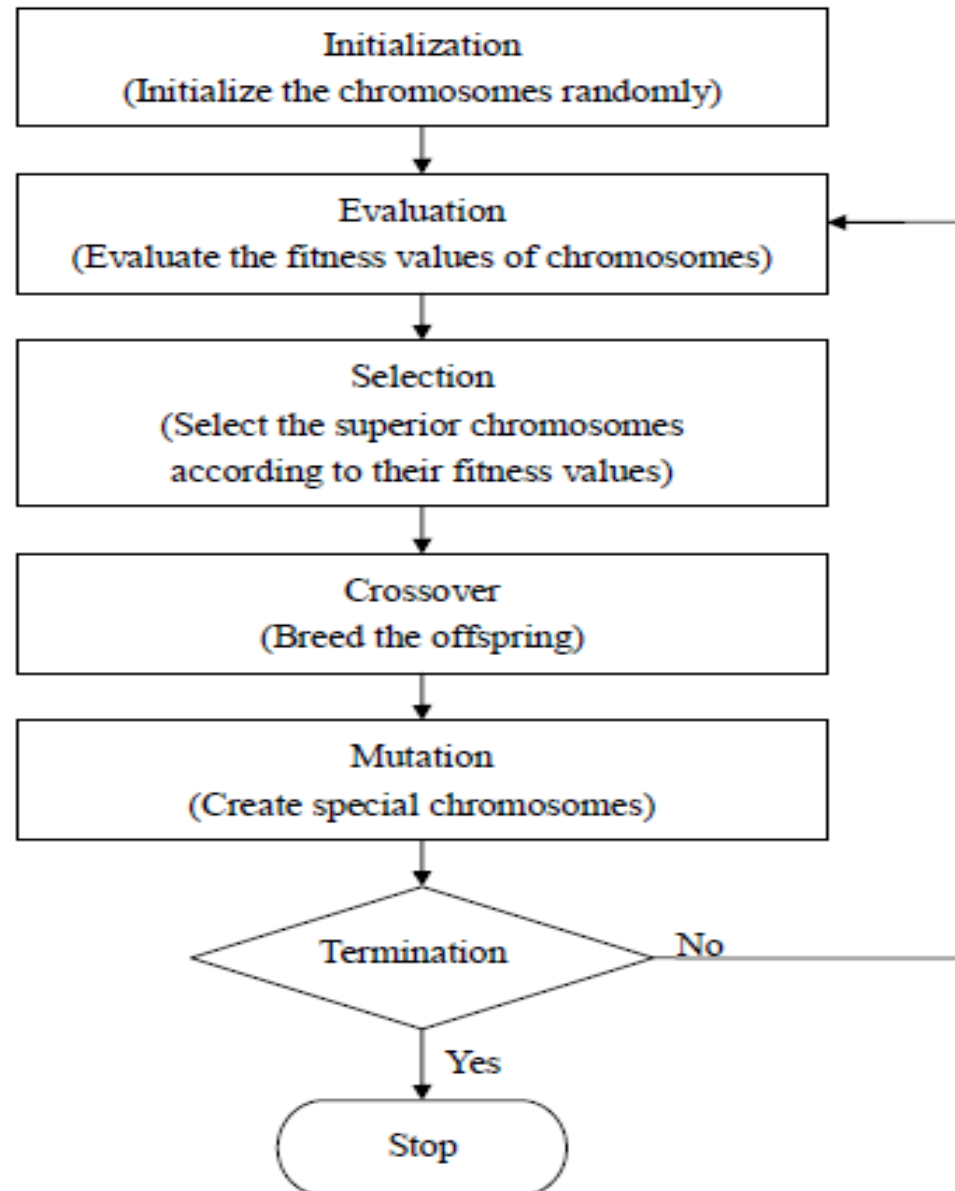
$$\text{Minimize } \lambda \left[\sum_{t=1}^T (r_t - \bar{r})^2 / T \right] - (1 - \lambda) \bar{r} - \theta \left[\left(\sum_{t=1}^T (r_t - \bar{r})^3 / T \right) / \left(\sum_{t=1}^T (r_t - \bar{r})^2 / T \right)^{3/2} \right]$$

Subject to Eqs. (17)–(23)

Θ : weighing factor for skewness

- Balances variance, expected return and skewness at the same time.

Genetic algorithm



Population size = 100

Fitness function(mean-variance model) :

$$f = \lambda \left[\sum_{i \in Q} \sum_{j \in Q} w_i w_j \sigma_{ij} \right] - (1 - \lambda) \left[\sum_{i \in Q} w_i \mu_i \right]$$

Chromosome: two distinct part

Part 1:

Set Q of K distinct assets

Part2:

K real numbers S_i ($0 \leq S_i \leq 1$)

S_i : the share of the free portfolio proportion

$\sum_{j \in Q} \varepsilon_j$: the total portfolio is already accounted

$$w_i = \varepsilon_i + \left(S_i / \sum_{j \in Q} S_j \right) \left(1 - \sum_{j \in Q} \varepsilon_j \right)$$

Uniform crossover

Mutation: 0.9 or 1.1 (chosen with equal probability)

Experiment

Market(January 2004 ~ December 2006):

- HANG SENG(Hong Kong) : 33 assets
- FTSE(U.K.) : 93 assets
- S&P 100(U.S.A.) : 99 assets

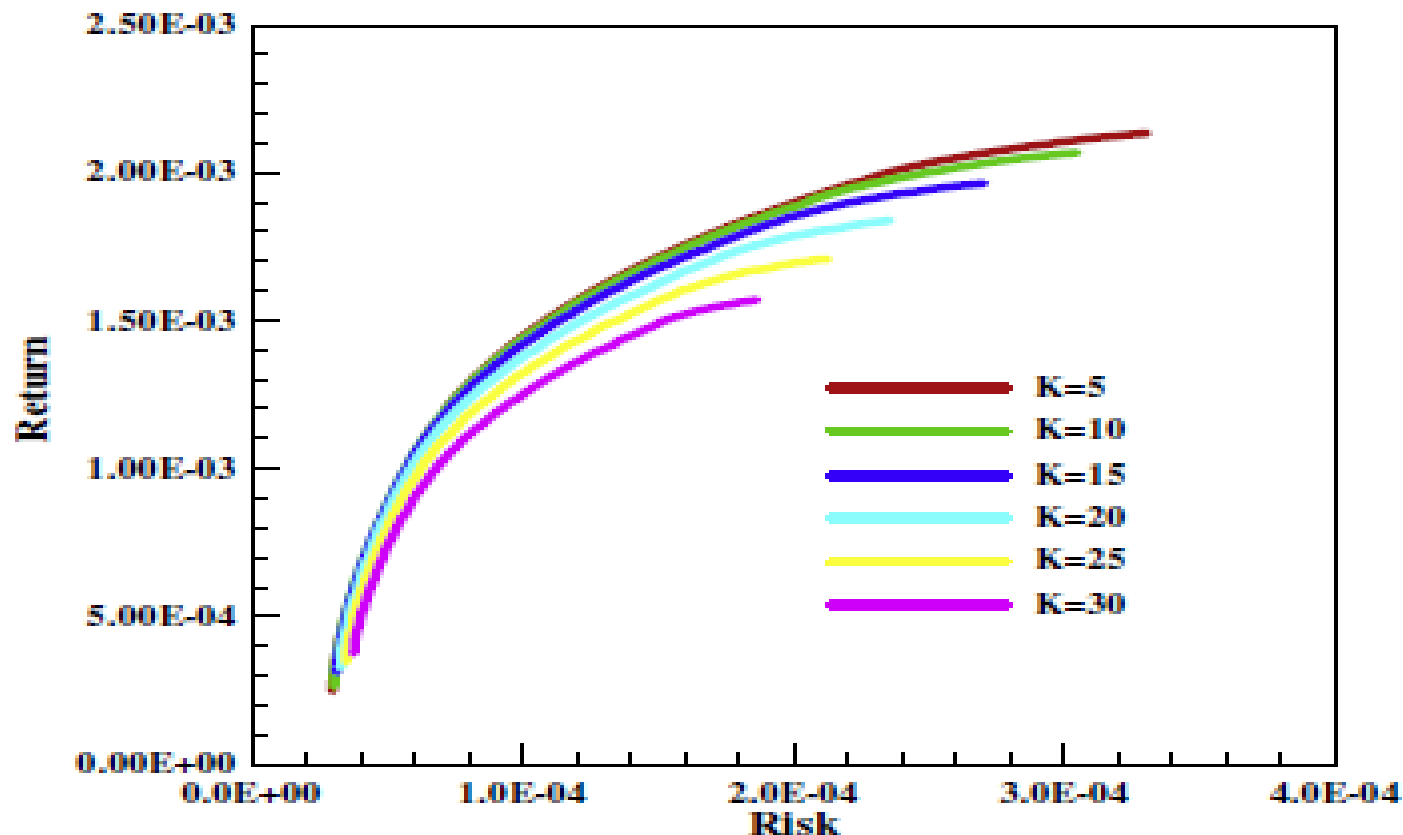


Fig. 1. CCEF of mean-variance model for HS data.

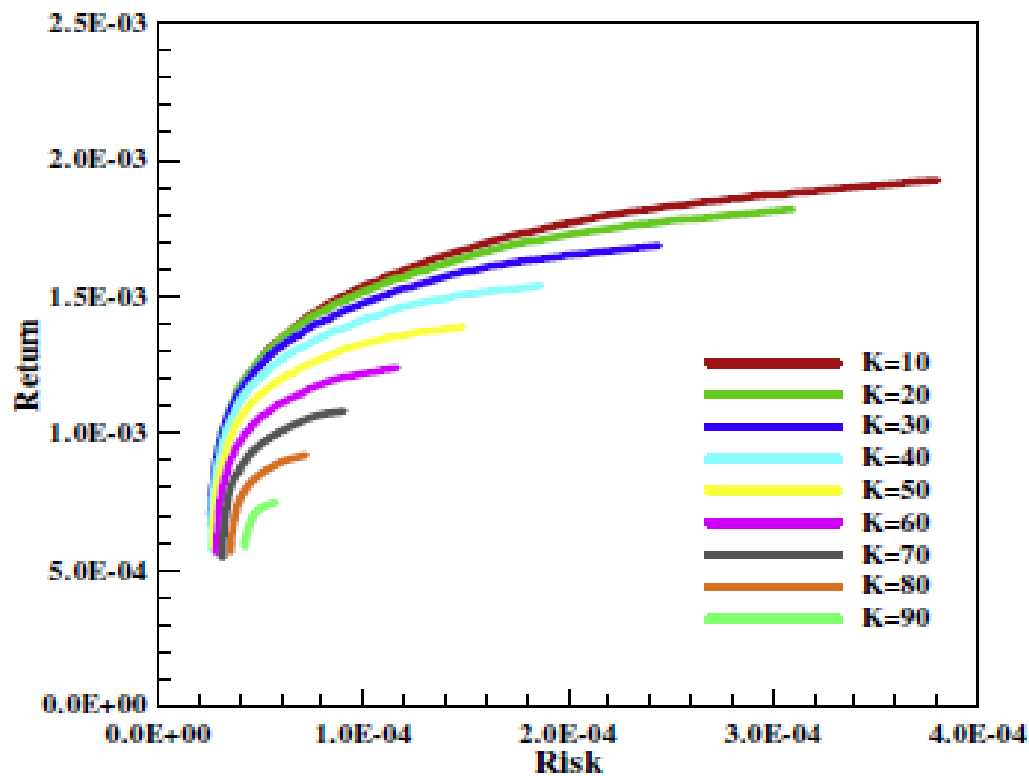


Fig. 2. CCEF of mean-variance model for FTSE data.

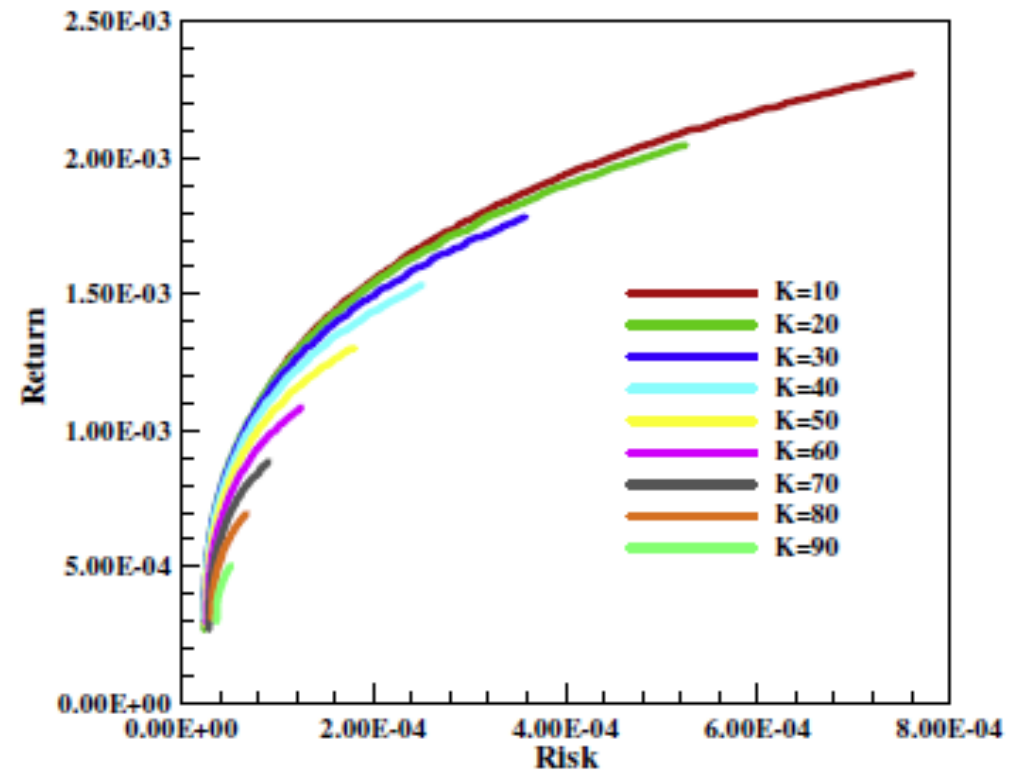


Fig. 3. CCEF of mean-variance model for S&P 100 data.

Cardinality constrained efficient frontier with a K values above **one third of total** is obviously dominated by those with relatively less K values.

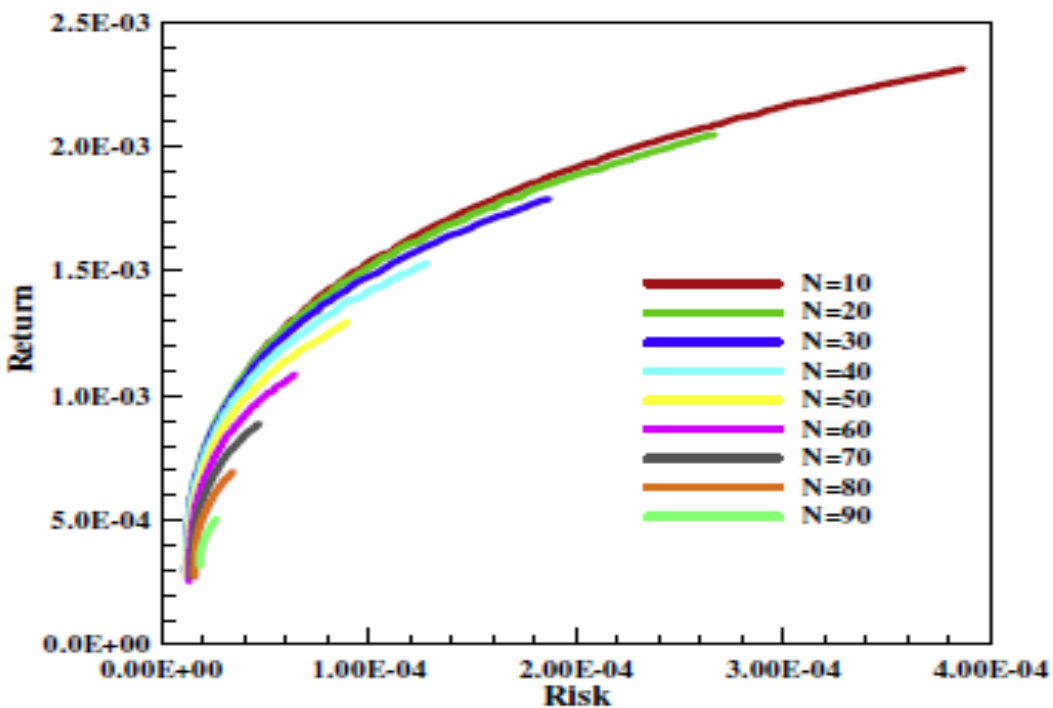


Fig. 4. CCEF of semi-variance model for S&P 100 data.

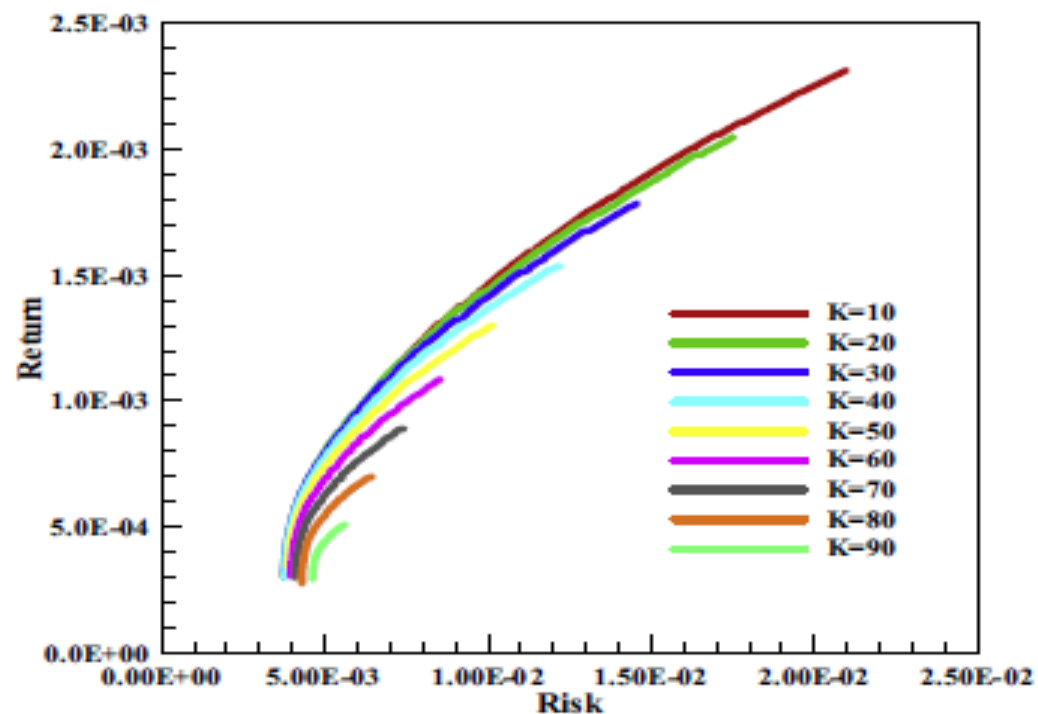


Fig. 5. CCEF of mean absolute deviation model for S&P 100 data.

Cardinality constrained efficient frontier for different models
 have similar results as those in mean-variance model.

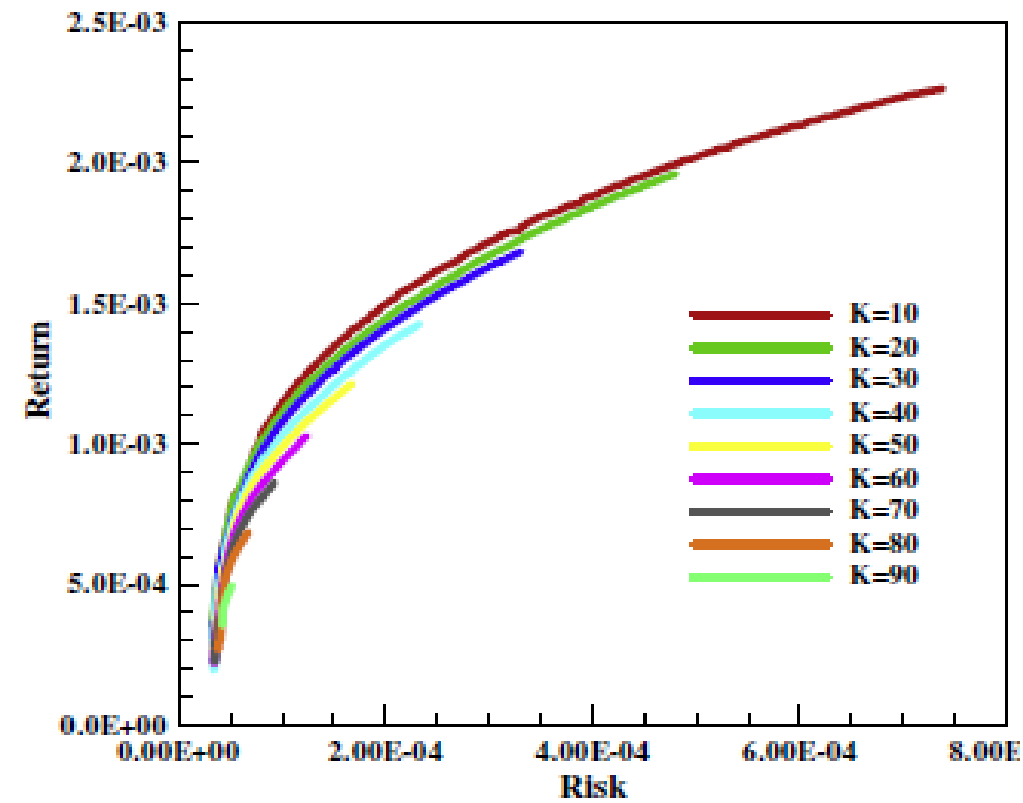


Fig. 6. CCEF of variance with skewness model for S&P 100 data.

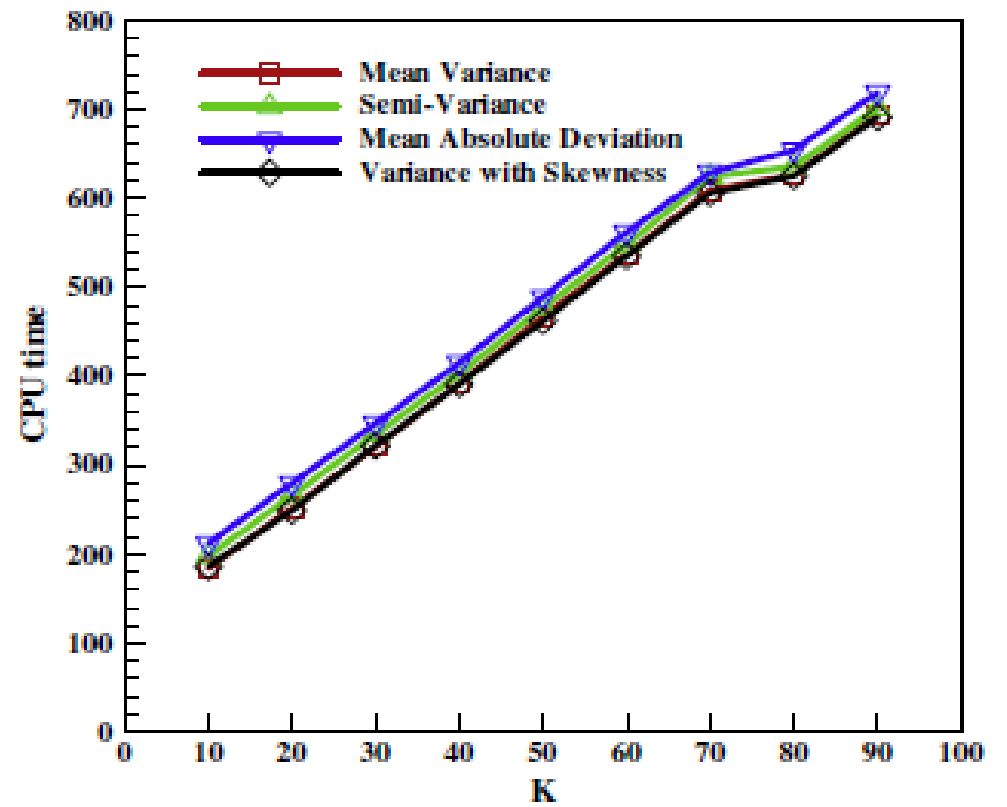


Fig. 7. CPU time (s) for various values of K in different risk models.

Thank you !!